V-L-51b
July 2, 1964

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May 20, 1964 and
Fed. Spec. V-L-51
November 7, 1946

FEDERAL SPECIFICATION

LACES, FOOTWEAR, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton laces for use as closures for footwear and other items.

1.2 Classification.

1.2.1 *Types and classes.* The cotton laces covered by this specification shall be furnished in the following types and classes, as specified (see 6.2).

Type I.—Mercerized.

Type II.—Soft finish.

Type III.—Glazed finish.

Type IV.—Waxed finish.

1.2.1.1 *Classes.* Cotton shoe laces shall be furnished in the classes indicated in table I for each type, as specified in the invitation for bids. The classes are based on differences in yarn size, number of carriers, and number of picks per inch.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

C-B-191—Beeswax; Technical Grade.

UU-P-268—Paper, Kraft, Untreated Wrapping.

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

VV-W-95—Wax, Paraffin, Technical.

CCC-T-191—Textile Test Methods.

JJJ-W-141—Wax, Carnauba.

PPP-B-636—Box, Fiberboard.

PPP-T-60 — Tape, Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products, Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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books and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples.

3.1.1 *Guide sample.* Guide samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from this specification may appear in the sample, in which case the specification shall govern.

3.1.2 *Standard sample.* When a standard sample is established, the finished laces shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.1.3 *Preproduction sample approval.* When specified (see 6.2), before production is commenced 20 finished laces shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.2 Material.

3.2.1 *Cotton.* The cotton used in the manufacture of the yarn for all types of laces shall be cleaned, carded, drawn, spun and twisted into yarn in the size and ply specified in table I. The yarn for type I laces shall be mercerized.

3.2.2 *Tips.* Unless otherwise specified (see 6.2), the tips shall be made from nylon sheet or film having a thickness of 0.013 to 0.016 inch. The color of the tips shall be black for black laces and clear for all other color laces.

3.2.3 *Wax.*

3.2.3.1 *Beeswax.* The beeswax shall conform to C-B-191.

3.2.3.2 *Cerese wax.* The cerese wax shall be a yellow microcrystalline wax having a melting point of 168° to 170°F. when tested as specified in 4.3.1, and having a specific gravity of 0.92 to 0.94 at 60°F. when tested as specified in 4.3.1.

3.2.3.3. *Carnauba wax.* The carnauba wax shall conform to grade B, type 3 of JJJ-W-141.

3.2.3.4 *Paraffin wax.* The paraffin wax shall conform to type I, grade B of VV-W-95 and shall have a melting point of 130° to 152°F.

3.2.3.5 *Ceresin wax.* The ceresin wax shall have a melting point of 174° to 176°F. when tested as specified in 4.3.1.

3.3 Color. The color of the finished laces shall be as specified (see 6.2). Unless specifically authorized, the use of sulfur dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed laces show no more free sulfur or sulfide sulfur than the standard sample (see 6.3), when tested as specified in 4.3.2.2.

3.3.1 *Matching.* The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500°Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

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TABLE I.—*Physical requirements*

Type	Class	Width ± 1/32"	Yarn size, ply	Ends per carrier	Number of carriers	Picks per inch	Breaking strength
		Tolerance, inches	*Minimum*	*Minimum*	*Minimum*	*Minimum*	*Minimum, pounds*
I	A	7/32	54/2	2	44	32	70
I	B	7/32	50/2	1	64	38	70
I	C	7/32	40/2	2	44	26	70
II	A	8/32	20/2	2	32	16	120
II	B	11/32	20/2	2	44	16	150
III	A	8/32	20/2	2	32	16	120
III	B	11/32	20/2	2	44	16	150
IV	A	7/32	24/2	2	40	22	110
IV	B	7/32	26/2	2	44	22	110
IV	C	7/32	30/2	2	48	26	110
IV	D	10/32	28/2 to 30/2	2	44	16	110

3.3.2 *Colorfastness.* The dyed laces shall show fastness to water equal to or better than the standard sample when tested as specified in 4.3.2.2. When no standard sample has been established, the dyed laces shall show good fastness to water when tested as specified in 4.3.2.2.

3.3.3 *pH.* The pH value of the dyed laces shall be not more than 8.0 nor less than 4.0 when tested as specified in 4.3.2.2.

3.4 **Braid.** The braid for the laces shall be tubular braided without a core using two over and two under braiding. The braid shall be pressed flat and shall conform to the requirements in table I when tested as specified in 4.3.2.2.

3.5 **Finish.**

3.5.1 *Type I.* The type I lace shall have a smooth and silky lustre.

3.5.2 *Type II.* The type II lace shall have a soft finish.

3.5.3 *Type III.* The type III lace shall be glazed and have a smooth glossy finish.

3.5.4 *Type IV.* The type IV lace shall be waxed by immersing in a molten wax solution as specified in 3.5.4.1. The wax shall thoroughly saturate and shall be evenly distributed throughout the lace. After passing through the wax solution, the lace exclusive of tip shall increase in weight not less than 17.5 nor more than 33.5 percent based on the finished weight of lace when tested as specified in 4.3.2.2. The finished lace shall contain a minimum of 0.04 percent to a maximum of 0.09 percent metallic copper as copper-8-quinolinolate when tested as specified in 4.3.2.2.

3.5.4.1 *Wax solution.* The wax solution shall consist of the following waxes in the amount specified in table II to which there shall be added sufficient copper-8-quinolinolate in order to conform to requirements specified in 3.5.4.

TABLE II.

	Percent:
Beeswax	5.25
Cerese wax	5.25
Carnauba wax	36.85
Paraffin wax	36.85
Ceresin wax	15.80
Total	100.00

3.6 **Tipping.** Both ends of the lace shall be provided with a firmly attached and secured tip made from the material specified in 3.2.2. Tips shall be 3/4 ± 1/16 inch long.

3.6.1 *Gripping strength.* The tips shall have a minimum gripping strength of 20 pounds when tested as specified in 4.3.2.2.

3.6.2 *Compression strength.* The tips shall not be cracked, dislodged from their posi-

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tion, or peeled from the lace when tested as specified in 4.3.2.2.

3.7 Length. Laces shall be furnished in lengths, including tips, as specified (see 6.2). The laces shall be cut so as to insure the specified length when tested as specified in 4.3.2.2. Laces under 40 inches may have a length tolerance of plus 5 percent, and laces 40 inches and longer may have a length tolerance of plus 3 percent.

3.8 Workmanship. The finished laces shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When required, the preproduction sample submitted in accordance with 3.1.3 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical characteristics in accordance with 4.3.2.2 to the extent applicable.

4.3 Inspection. Inspection shall be in accordance with the provisions of MIL-STD-105 except where otherwise indicated.

4.3.1 *Testing of components.* In addition to the quality assurance provisions of subsidiary specifications and drawings, testing shall be performed on the components listed in table III for the characteristics shown herein. The inspection level for each component shall be level S-1 and the acceptable quality level (AQL) for each component shall be 4.0 test failures per hundred units. All requirements are applicable to the sample unit.

4.3.2 *Inspection of the end item.*

4.3.2.1 *Examination of the end item.* The laces shall be examined for defects in workmanship, material, and overall length of tips.

TABLE III.—*Component tests*

Component and lot expressed in terms of	Characteristic	Rqmt. par.	Test method	No. det. per sample unit	Results reported as—	Sample unit
Nylon sheet (10 sheets)	Thickness	3.2.2	Gauge	3	Avg. of 3 det. to the nearest 0.001 inch	6 x 1-1/2 inch piece or equivalent
Cerese wax (1 pound)	Melting point	3.2.3.2	Method 1401 of Fed. Test Method Std. No. 791	1	Pass or fail	1/4 (lb.)
	Specific gravity	3.2.3.2	4.4.7	1	Pass or fail	
Ceresin wax (1 pound)	Melting point	3.2.3.5	Method 1401 of Fed. Test Method Std. No. 791	1	Pass or fail	1/4 (lb.)
Wax solution	Composition	[1] 3.5.4.1	Standard Comm'l	—	—	—

[1] The supplier shall submit a certificate of compliance for this characteristic.

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Defects found during these examinations shall be classified in accordance with 4.3.2.1.1 and 4.3.2.1.2. The sample unit for these examinations shall be one completely finished lace and the lot size shall be expressed in units of one lace each.

4.3.2.1.1 *Examination of laces for defects in material and workmanship.* The inspection level for this examination shall be level I of MIL-STD-105 and the acceptable quality level (AQL) shall be 1.0 defects per hundred units.

Examine	Defect
Material	Not fabricated of the specified material.
	Any broken thread.
	Frayed.
	Rub or abrasion marks.
	Hole, cut or tear.
	Unevenly braided resulting in open place or break in continuity of braid.
	Pulled thread or uneven tension resulting in uneven width.
	Other than tubular braided without a core
	Not flat for a distance of 2 inches or more.
	Slub, knot, or abruptly thickened place which may affect utility.
	Color other than specified.
	Visibly off shade, i.e., not within established tolerances.
Material (cont'd.)	Spot, stain, or other discoloration affecting appearance.
	Objectionable odor.
	Finish not as specified or not good match to standard or approved sample.
Tipping	Any tip missing, damaged, malformed or insecurely attached.
	Tipping of braid not neatly accomplished.
	End of tip projects beyond the braid more than 1/32 inch.
	End of braid projects beyond the tip end more than 1/32 inch.

4.3.2.1.2 *Examination for overall length of tip ends.* Any overall length of tip exceeding specified tolerances indicated in 3.7 shall be classified a defect. The inspection level for this examination shall be S-3 of MIL-STD-105 and the acceptable quality level (AQL), shall be 1.0 defects per hundred units.

4.3.2.2 *Testing of the end item.* Testing of the completely fabricated laces shall be performed in accordance with table IV and CCC-T-191 for characteristics shown therein. The inspection level shall be S-1 of MIL-STD-105 and the acceptable quality level

TABLE IV.—*End item tests*

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Cotton identification	3.2.1	1200	—	—
Sulfur content	3.3	4.4.8	1	Pass or fail
Colorfastness to water	3.3.2	4630	—	—
pH value	3.3.3	2811	—	—
Weave	3.4	Visual	1	Pass or fail
Width, overall	3.4	4.4.1	3	To the nearest 1/32 inch
Yarn size	3.4	4020	—	—
Yarn ply	3.4	Visual	1	Number of plies
Ends per carrier	3.4	Visual	1	Number of ends
Carriers	3.4	Visual	1	Number of carriers
Picks per inch	3.4	Visual	1	Number of picks
Breaking strength	3.4	4.4.3	—	—
Wax content	3.5.4	4.4.6	3	To the nearest 0.1 percent
Copper content, percent	3.5.4	2050 or 2051	—	—
Gripping strength (tips)	3.6.1	4.4.5	—	—
Compression strength (tips)	3.6.2	4.4.4	1	Pass or fail
Length, overall	3.7	4.4.2	1	To the nearest 1/16 inch

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(AQL) shall be 2.5 test failures per hundred units. The sample unit for tests shall be twenty completely fabricated laces. Requirements are applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table IV they shall be as required by the referenced test method. The lot size shall be expressed in units of one lace each.

4.3.3 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Examination shall be in accordance with the provisions of MIL-STD-105. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or damaged affecting serviceability.
	Any component not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Content	Number of laces per bundle is more or less than required.[1]
Weight	Weight of the contents of container exceeds the maximum requirement.

[1] One bundle from each shipping container in the sample shall be examined.

4.4 Tests.

4.4.1 *Width.* The width shall be determined at 3 points at least 9 inches apart by a steel rule calibrated in thirty-seconds of an inch, or by a caliper type gauge and the results averaged. The measurements shall be taken with the test lace suspended vertically under a tension of 3 ounces.

4.4.2 *Length.* The finished lace shall not be stretched or tensioned prior to the length test. The finished lace shall be suspended vertically under a tension of 3 ounces and then measured from end to end including the tips.

4.4.3 *Breaking strength.* The breaking strength of the laces shall be determined as specified in method 4102 of CCC-T-191, except that the clamps shall be not less than 2 inches in diameter and the distance between the center of the clamps at the start of the test shall be 6 ± 1/16 inches. When the lace length is less than 24 inches, a flat type jaw shall be used in lieu of a drum type jaw.

4.4.4 *Compression strength.* The tip shall be placed between flat blocks of steel and subjected to a gradually applied crushing load of 1000 pounds.

4.4.5 *Gripping strength of tips.*

4.4.5.1 *Test specimen.* The specimen shall be 1/2 length of a shoe lace. (Two determinations may be made from one complete shoe lace.)

4.4.5.2 *Apparatus.*

4.4.5.2.1 *Testing machine.* A testing machine as described in method 5100 except a slotted metal plate, as shown in figure 1, shall be used in connection with the top jaw assembly and the distance between the upper jaw (slotted metal plate) and the lower jaws or clamps shall be 4 ± 1/16 inches. The lower jaw shall be described in method 5100 or 4102 of CCC-T-191.

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4.4.5.2.2 *Metal plate* (*figure* 1). A rigid metal plate of the approximate dimensions 5 inches long by 1-1/2 inches wide, containing a 4-inch tapered slot, centered widthwise and running lengthwise from an inside dimension of 1/32 inch in width, increasing to a width of 1/4 inch at the outside edge. The slot shall be smoothly finished. It is not required that the plate be of a single piece of metal. The slot may be formed by a clamped or bolted assembly of two metal bars or plates properly spaced apart. The plate shall be supported in a horizontal position in or on top jaw of the apparatus.

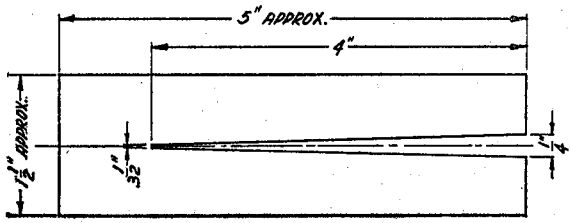


FIGURE I.—Slotted metal plate.

4.4.5.3 *Procedure.* The lace shall be passed upward, tip first, through the jaw assembly and through the wide end of the slot until the tip clears the plate and then moved horizontally toward the narrow end of the slot until the end of the tip rests on top of the metal plate. The free end of the specimen (lace) shall be fastened in the lower jaw or clamp. The machine shall be started, with the pawl in contact with the ratchet and the maximum resistance in displacing the tip shall be recorded. The force shall be applied to the specimen at such a rate that the pulling clamp will travel at a uniform speed of 12 ± 0.5 inches per minute. After the tip has been forced from the lace, the resultant force shall be read from the dial, scale or chart and the value recorded to the nearest pound. If the lace tip slips through the slot without its being removed from the lace or if for any reason attributed to faulty technique, an individual measurement falls markedly below the average test for the sample unit, such result shall be discarded and another specimen shall be tested. Buckling of the tip material at the base of the tip without the tip being removed from the lace shall not be considered as a failure of the tip.

4.4.5.4 *Report.* Ten specimens shall be tested from each sample unit. The gripping strength of the sample unit shall be the average of the results and shall be reported to the nearest 1.0 pounds.

4.4.6 *Wax content.* Unless otherwise specified, petroleum ether boiling below 80°C. shall be used as a solvent. The solvent shall contain not more than 1 mg. residue per 100 ml. when dried to constant weight at 98° to 100°C.

4.4.6.1 *Apparatus.* The apparatus shall consist of a soxhlet apparatus, extraction thimble, electric hot plate, and drying chamber.

4.4.6.2 *Procedure.* The specimen shall be accurately weighed and transferred to the extraction thimble. The thimble shall be placed in the extraction chamber of the soxhlet. The soxhlet flask accurately weighed, shall be approximately three fourths filled with the specified solvent and the apparatus assembled. The lace shall be subjected to not less than 20 extractions with the hot plate adjusted so that the duration of each extraction period shall be 15 ± 3 minutes. The flask shall be removed and the solvent evaporated on a steam bath. The residue shall be dried to constant weight in an oven at 98° to 100°C., placed in a dessicator to cool, and then weighed. The percentage of wax shall be calculated from the increased weight of the flask. The wax shall be reported in percent.

4.4.6.3 *Report.* Separate determinations shall be made on three different laces and all determination results shall be within the minimum and maximum requirements specified.

4.4.7 *Determination of specific gravity of ceresin wax.* Melt approximately 2 grams of the sample in a porcelain crucible at a tem-

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perature of 100° to 105°C. Allow to cool to room temperature slowly. Remove the solidified wax from the crucible, warming the crucible slightly, if necessary, and attach a tared silk thread to suspend the wax during weighing. Store the sample for two hours at a temperature of approximately 15°C. and then determine the weight of the sample suspended first in air and then in alcohol at a temperature of 25.0°C. Determine the specific gravity (25°C/25°C) of the alcohol by means of a hydrometer or Westphal balance.

Calculate the specific gravity using the following equation:

$$\text{Specific gravity} = \frac{W_1 \times (\text{sp. gr. of alcohol})}{W_1 \times W_2}$$

Where W_1 = weight of sample in air.
and W_2 = weight of sample in alcohol.

4.4.8 *Test for sulfur content.*

4.4.8.1 *Solution (acid stannous chloride).* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.8.2 *Procedure.* One to 1.5 grams of the finished lace, cut into small lengths, shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous chloride solution. Place a filter paper which has been moistened with one drop of 5.0 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.8.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Five gross of laces of one type, class, length, and color only shall be put-up in a compact bundle. Prior to bundling, the laces shall be grouped full length in one-gross lots and securely tied together. Each bundle of laces shall be completely wrapped in 60-pound basis weight kraft paper conforming to grade B of UU-P-268. The wrap shall be securely sealed with 2-inch minimum width gummed paper tape conforming to class 2 of UU-T-111.

5.1.2 *Level C.* Laces shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 *Level A.* Laces of one type, class, length and color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c, or V3s, style RSC of PPP-B-636. Fiberboard containers shall have all seams, corners, and manufacturer's joint sealed by the application of 3-inch minimum width olive drab color cloth tape conforming to type III, class 1 of PPP-T-60. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

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5.2.2 *Level B.* Laces of one type, class, length, and color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. The minimum bursting strength of the fiberboard shall be 275 p.s.i. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.3 *Level C.* Laces, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The cotton laces covered by this specification are intended for use in footwear and other items.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type, class and length required (see 1.2 and 3.7).
(*c*) When preproduction sample is required (see 3.1.3).
(*d*) Tip material if other than specified (see 3.2.2).
(*e*) Color required (see 3.3).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 **Sample.** For access to guide and standard samples, address the procuring office issuing the invitation for bids.

6.4 **Civil agency procurement.** When level B packaging is specified for civil agency procurement, the requirements in 5.1.1 shall apply.

6.5 This specification is a revision of Federal Specification V-L-51 dated November 7, 1946. The comparison of the types and classes are as follows:

V-L-51a	*V-L-51*
Type I, Class A	Type I, Class A Type I, Class B
Type I, Class B	—
Type I, Class C	Type I, Class C
—	Type I, Class D (Deleted)
Type II, Class A	Type II, Class E
Type II, Class B	Type II, Class F
Type III, Class A	Type III, Class G
Type III, Class B	Type III, Class H
Type IV, Class A	Type IV, Class I
Type IV, Class B	Type IV, Class J
Type IV, Class C	Type IV, Class K
Type IV, Class D	Type IV, Class L

6.6 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Laces, Shoe:
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

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Motor:

Laces, braided shoe:
Truckload minimum weight 18,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—GL

Navy—S&A

Air Force—SEG

Interested activities:

Review

Navy—S&A

Air Force—MAA

User

Army—MD

Air Force—SEG

Preparing activity

Army—GL

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